

Multiple Choice

1. **Answer:** A

Options: A) By overwriting the return address to point to the location of that code; B) By writing directly to the instruction pointer register the address of the code; C) By writing directly to %eax the address of the code; D) By changing the name of the running executable, stored on the stack

Note: Correct option: A - By overwriting the return address to point to the location of that code

2. **Answer:** A

Options: A) It generates each different input by modifying a prior input; B) It works by making small mutations to the target program to induce faults; C) Each input is mutation that follows a given grammar; D) It only makes sense for file-based fuzzing, not network-based fuzzing

Note: Correct option: A - It generates each different input by modifying a prior input

3. **Answer:** B

Options: A) No, it cannot be done; B) Yes, just plug the GGM PRF into the Luby-Rackoff theorem; C) It depends on the underlying PRG; D) Option text

Note: Correct option: B - Yes, just plug the GGM PRF into the Luby-Rackoff theorem

4. **Answer:** D

Options: A) Opera browser; B) Firefox; C) Chrome; D) Tor browser

Note: Correct option: D - Tor browser

5. **Answer:** D

Options: A) Yes, if the PRG is really "secure"; B) No, there are no ciphers with perfect secrecy; C) Yes, every cipher has perfect secrecy; D) No, since the key is shorter than the message

Note: Correct option: D - No, since the key is shorter than the message

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'SMS Trojan'.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Yes, the key is $k = m \text{ xor } c$.'

Long Form Response

11. **Answer:** `def find_demlo(s): | return res`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def find_adverb_position(text): | return (m.start(), m.end(), m.group(0))`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key